

$$T(n) = \begin{cases} 1, & \text{se } n \leq 2; \\ 2 \cdot T(\sqrt[n]{n}) + \log(2n), & \text{altrimenti.} \end{cases}$$

LIVELLO		CONTRIBUTI SINGOLI	NUM RAMI	CONTRIBUTO TOTALE
0	$T(n) = 2 \cdot T\left(n^{\frac{1}{2}}\right) + \log_2(2n)$	$\log_2(2n)$ $= 1 + \log_2(n)$	1	$\log_2(2n) =$ $= 1 + \log_2(n)$
1	$T\left(n^{\frac{1}{2}}\right) = 2 \cdot T\left(n^{\frac{1}{4}}\right) + \log_2\left(2 n^{\frac{1}{2}}\right)$	$\log_2\left(2 n^{\frac{1}{2}}\right)$ $= 2 + \frac{1}{2} \log_2(n)$	2	$2 \cdot \log_2\left(2 n^{\frac{1}{2}}\right)$ $= 2 \cdot (\log_2(2) +$ $\log_2\left(n^{\frac{1}{2}}\right)) =$ $= 2 + \frac{1}{2} \log_2(n)$
2	$T\left(n^{\frac{1}{4}}\right) = 2 \cdot T\left(n^{\frac{1}{8}}\right) + \log_2\left(2 n^{\frac{1}{4}}\right)$	$\log_2\left(2 n^{\frac{1}{4}}\right)$ $= 1 + \frac{1}{4} \log_2(n)$	$2 \cdot 2 = 4$	$4 \cdot \left(1 + \frac{1}{4} \log_2(n)\right)$ $= 4 + \frac{1}{4} \log_2(n)$

POTREI ANDARE AVANTI, MA NON CREDO SIA NECESSARIO, IL CONTRIBUTO PER LIVELLO È:

$$2^i + \frac{1}{2^i} \log_2(n)$$

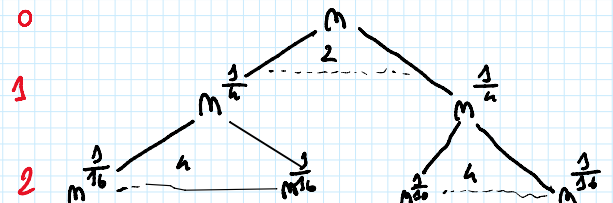
MENTRE L'ALTEZZA DELL'ALBERO SEGUE LA REGOLA

$$n^{\left(\frac{1}{2^i}\right)^i}$$

RICAVIAMOCI L'ALTEZZA DELL'ALBERO H SEGUENDO IL CASO PEGGIORE

$$n^{\frac{1}{2^h}} = 2 \Rightarrow \log_2\left(n^{\left(\frac{1}{2^h}\right)^h}\right) = \log_2 2 \Rightarrow$$

$$\left(\frac{1}{2^h}\right)^h \log_2(n) = 1 \Rightarrow \frac{1}{2^h} \log_2(n) = 1 \Rightarrow \log_2(n) = 2^h$$



$$\log_2(\log_2(n)) \cdot \log_2(2^{2^h}) \Rightarrow \log_2(\log_2(n)) = 2h \Rightarrow h = \frac{\log_2(\log_2(n))}{2}$$

CALCOLO DELLA SOMMATORIA

$$\sum_{i=0}^h 2^i + \sum_{i=0}^h \frac{1}{2^i} \log_2(n) =$$

$$\sum_{i=0}^h 2^i = 2^{h+1} \text{ QUINDI SERIE GEOMETRICA COMPLETA} \Rightarrow \frac{2^{h+1} - 1}{2 - 1} = 2^{h+1} - 1$$

$$\Rightarrow 2^{\frac{\log_2(\log_2(n))}{2} + 1} - 1$$

$$\log_2(n) \sum_{i=0}^h \frac{1}{2^i} = \frac{1}{2^h} \text{ E' COMPRESO TRA } 0 \text{ E } 1 \Rightarrow \frac{1}{1 - \frac{1}{2}} = \frac{1}{\frac{1}{2}} = 2 \log_2(n)$$

SI FRUITO LA PROPRIETA'

$$m = a^{\log_a(m)}$$

$$2^{\frac{\log_2(\log_2(n))}{2} + 1} - 1 + 2 \log_2(n)$$

$$\sqrt{2^{\log_2(\log_2(n))}} \cdot 2 - 1 + 2 \log_2(n) \Rightarrow \sqrt{\log_2(n)} \cdot 2 - 1 + 2 \log_2(n)$$

$$\Theta(\log_2(n))$$

2. Si scriva un **algoritmo iterativo** che **simuli precisamente** l'algoritmo ricorsivo di seguito riportato, dove Z è una funzione esterna non meglio specificata.

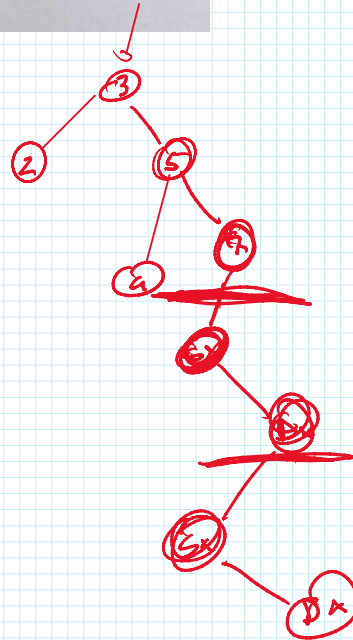
```

1  if  $T = \text{Nil}$  then
2    return  $Z(0, h)$ 
3  else
4     $a \leftarrow 0$ 
5    if  $T \rightarrow \text{key} \equiv 0 \bmod 2$  then
6       $\lfloor a \leftarrow a + \text{Algoritmo}(T \rightarrow dx, 2 \cdot h)$ 
7    if  $T \rightarrow \text{key} \equiv 1 \bmod 3$  then
8       $\lfloor a \leftarrow a - \text{Algoritmo}(T \rightarrow sx, 3 \cdot h)$ 
9    return  $Z(T \rightarrow \text{key}, a)$ 

```

```
st_t = st_h = st_a = NULL
cH = h
cT = T
start = true
last = NULL
```

```
start = false // RIPETITIVO
```



```
ret = Z(cT->key, a)
st_t = pop(st_t)
st_h = pop(st_h)
st_a = pop(st_a)
last = cT
```

```
return ret
```

```
1  if  $j - i \geq 1$  then
2       $y = \text{Rand}() \% 2$ 
3      if  $(x = 1)$  then
4           $ret = \text{ALGORITMO}(A, \frac{(i+j)}{2} + 1, j, y)$ 
5          if  $ret \% 2 = 0$  then
6               $ret = \text{ALGORITMO}(A, i, \frac{(i+j)}{2}, 1 - y)$ 
7          else
8               $ret = \text{ALGORITMO}(A, i, \frac{(i+j)}{2}, y)$ 
9          if  $ret \% 2 = 1$  then
10              $ret = ret + \text{ALGORITMO}(A, \frac{(i+j)}{2} + 1, j, 1 - y)$ 
11 else
12      $ret = A[i]$ 
```

$$T(n) = \begin{cases} 1, & \text{se } n \leq 27; \\ 3n^2 \cdot T(\sqrt[3]{n}) + 2n^3, & \text{altrimenti.} \end{cases}$$

LIVELLO		CONTRIBUTI SINGOLI	NUM RAMI	CONTRIBUTO TOTALE
0	$T(n) =$ $3n^2 \cdot T(\sqrt[3]{n}) + 2n^3,$ $\downarrow T(n^{1/3})$	$2n^3$	1	$2n^3$
1	$T(n^{1/3}) = 3 \cdot (n^{1/3})^2 \cdot T((n^{1/3})^{1/3}) +$ $2 \cdot (n^{1/3})^3 =$ $= 3n^{2/3} \cdot T(n^{1/9}) + 2n$	$2n$	$3n^2$	$2n \cdot 3n^2$ $6n^3$
2	$T(n^{1/9}) = 3 \cdot (n^{1/9})^2 \cdot T((n^{1/9})^{1/3}) +$ $2 \cdot (n^{1/9})^3 =$ $3n^{2/9} \cdot T(n^{1/27}) + 2 \cdot (n^{1/9})$	$2n^{1/3}$	$3n^{2/3} \cdot$ $\frac{3n^2}{3n^{2/3}}$ $3n^{4/3}$	$\frac{2n^{1/3} \cdot 3n^{2/3}}{3n^{4/3}} =$ $18n^3$

CASO PERMANENTE

$$n^{1/3} \leq 27$$

$$n^{1/3} = 27 \Rightarrow$$

CONTRIBUTO GENERALE
ASYMPTOTICO

$$2n^{3/4}$$

$$\log_3 n^{1/3^h} = \log_3 27 \Rightarrow \frac{1}{3^h} \cdot \log_3(n) = 3 \Rightarrow$$

$$\log_3(n) = 3 \cdot 3^h \Rightarrow \log_3(n) = 3^{h+1} \Rightarrow$$

$$\Rightarrow h+1 = \log_3 \log_3(n) \Rightarrow h = \log_3(\log_3(n)) - 1$$

SOLUZIONE PERFETTA X MOGAVERO

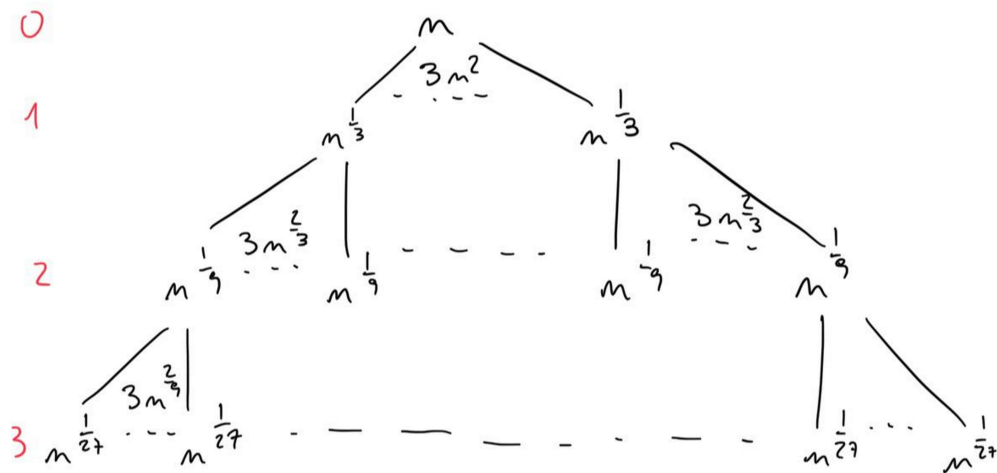
$$\begin{aligned}
 n^{1/3^i} &= 2^i \\
 \log_{2^i} n^{1/3^i} &= 1 \\
 \frac{1}{3^i} \cdot \log_{2^i}(n) &= 1 \\
 \log_{2^i}(n) &= 3^i \Rightarrow i = \log_3 \log_{2^i}(n)
 \end{aligned}$$

$$\begin{aligned}
 T(n) &= \sum_{i=0}^h 3^i \cdot 2n^3 = 2n^3 \sum_{i=0}^h 3^i = 2n^3 \cdot \left(\frac{3^{h+1} - 1}{3 - 1} \right) \\
 &= 2n^3 \cdot \frac{3^{h+1} - 1}{2} = n^3 \cdot \left(3^{(\log_3(\log_3(n)) - 1 + 1)} - 1 \right) = \\
 &= n^3 3^{\log_3(\log_3(n))} - n^3
 \end{aligned}$$

$$\Theta(n^3 \cdot \log_3(n))$$

SOLUZIONE LUCA

$$T(n) = \begin{cases} 1 & n \leq 27 \\ 3n^2 \cdot T(\sqrt[3]{n}) + 2n^3 & \text{altrimenti} \end{cases}$$



NUMERO NODI	LIVELLO
1	0
$3n^2$	1
$3n^2 \cdot 3n^{2/3}$	2
$3n^2 \cdot 3n^{2/3} \cdot 3n^{2/9}$	3

LIV.		CONTRIBUTO SINGOLO	CONTRIBUTO TOTALE
0	$3m^2 \cdot T(n^{\frac{1}{3}}) + 2m^3$	$2m^3$	$2m^3$
1	$3m^2 \left(T(n^{\frac{1}{3}}) \right) =$ $= 3m^2 \left(3m^{\frac{2}{3}} \cdot T(n^{\frac{1}{9}}) + 2m^{\frac{3}{3}} \right)$	$2m^{\frac{3}{3}}$	$3m^2 \cdot 2m =$ $= 6m^3$
2	$3m^{\frac{2}{3}} \cdot 3m^2 \cdot \left(T(n^{\frac{1}{9}}) \right) =$ $= \left(3m^{\frac{2}{3}} \cdot T(n^{\frac{1}{27}}) + 2m^{\frac{3}{3} \cdot \frac{1}{3}} \right)$ $\cdot 3m^{\frac{2}{3}} \cdot 3m^2$	$2m^{\frac{3}{9} \cdot \frac{1}{3}}$	$3m^2 \cdot 3m^{\frac{2}{3}} \cdot$ $2m^{\frac{1}{3}} =$ <u>$= 18m^3$</u>

POTREI ANDARE AVANTI, MA NON CREDO SIA NECESSARIO, IL CONTRIBUTO PER LIVELLO È:

$$3^i \cdot 2 \cdot m^3$$

MENTRE L'ALTEZZA DELL'ALBERO SEGUE LA REGOLA

$$m \left(\frac{1}{3} \right)^i$$

E POSSO RICAVERE LA i COSÌ:

$$n^{\left(\frac{1}{3}\right)^i} = 27 \Rightarrow \log_{27} n^{\left(\frac{1}{3}\right)^i} = \log_{27} 27^1$$

$$\left(\frac{1}{3}\right)^i \log_{27} n = 1$$

$$\log_{27} n = 3^i$$

$$\log_3 \log_{27} n = i$$

OPPURE

$$n^{\left(\frac{1}{3}\right)^i} = 27 \Rightarrow \log_3 n^{\left(\frac{1}{3}\right)^i} = \log_3 27^3$$

$$\left(\frac{1}{3}\right)^i \log_3 n = 3 \Rightarrow \log_3 n = 3 \cdot 3^i$$

$$\log_3 n = 3^{i+1} \Rightarrow \log_3 (\log_3 n) = \log_3 (3^{i+1})$$

$$\log_3 (\log_3 n) = i+1 \left(\log_3 3^1 \right)$$

$$i = \log_3 (\log_3 n) - 1$$

CALCOLO DELLA SOMMATORIA

$$\sum_{i=0}^h 2 \cdot 3^i \cdot n^3 \Rightarrow 2n^3 \sum_{i=0}^h 3^i \quad \begin{array}{l} 3 > 1, \text{ USO LA} \\ \text{FORMULA COMPLETA} \end{array}$$

$$2n^3 \left(\frac{3^{h+1} - 1}{3 - 1} \right) = 2n^3 \cdot \frac{1}{2} \cdot (3^{h+1} - 1) =$$

$$= n^3 \cdot (3^{h+1} - 1) \quad \begin{array}{l} \text{SOSTITUISCO } h \text{ COL} \\ \text{RICAVATO DI PRIMA} \end{array}$$

$$= n^3 \cdot \left(3^{\log_3 (\log_3 n) - 1 + 1} - 1 \right) =$$

$$= n^3 \cdot 3^{\log_3 (\log_3 n)} - n^3 =$$

$$= n^3 \cdot 3^{\log_3(\log_3 n)} - n^3 =$$

$$= n^3 \cdot (\log_3 n)^{\cancel{\log_3 3}} - n^3$$

$a^{\log_a(x)} = x$

$$\Theta(n^3 \log_3(n))$$

2

2. Si scriva un **algoritmo iterativo** che **simuli precisamente** l'algoritmo ricorsivo di seguito riportato, dove Z_l e Z_r sono due funzioni esterne non meglio specificate che soddisfano la seguente proprietà: $p < Z_l(A, p, s) < Z_r(A, p, s) \leq s$, quando $p + 1 < s$.

function Algoritmo(A, p, s)

```

1  if  $s \leq p + 1$  then
2    return 0
  else
3     $q \leftarrow Z_l(A, p, s)$ 
4     $r \leftarrow Z_r(A, p, s)$ 
5     $a \leftarrow \text{Algoritmo}(A, p, q)$ 
6     $a \leftarrow a - \text{Algoritmo}(A, q, r)$ 
7     $a \leftarrow a + \text{Algoritmo}(A, r, s)$ 
8    return  $a + (r - q)$ 

```

IF
C3
ELAB IF
C2
ELAB
C3

ALGO_IT(A, p, s)

$Cs = s$
 $Cp = p$
 $Start = \text{true}$
 $Last = \text{NIL}$
 $Ret = 0$
 $St_p = st_s = st_a = \text{NIL}$

While($start$ OR $st_p \neq \text{NIL}$) **DO**

If ($start$) **then**

If($cs \leq cp + 1$) **THEN**
 $Ret = 0$
 $Start = \text{false}$
 $Last = cp$

Else

// NUOVA CHIAMATA
 /* SIMULO FINO ALLA 1 CHIAMATA */

$q \leftarrow Z_l(A, cp, cs)$
 $r \leftarrow Z_l(A, cp, cs)$
 $St_p = \text{push}(st_p, cp)$
 $St_s = \text{push}(st_s, cs)$

else // VECCHIA CHIAMATA
 /* RECUPERO CONTESTO */

```
Cp = top(st_p)
Cs = top(st_s)
q <- Zl(A,cp,cs)
r <- Zl(A,cp,cs)
```

```
If(last = cp) THEN //torno 1 CHIAMATA DA CHIAMARE LA 2
```

```
  a <- ret
  push(st_a, a)
  Cp = q
  Cs = r
  Start = true
```

```
else // 2 CHIAMATA
```

```
  If(last = q) THEN // torno dalla 2 CHIAMATA da CHIAMARE LA 3
    a = top(st_a)
    A <- a - ret
    St_a = pop(st_a)
    Push(st_a, a)
    Cp = r
    Cs = cs
    Start = true
```

```
else // TORNO DALLA 3 CHIAMATA ///// LAST = R
```

```
  a = top(st_a)
  a <- a + ret
  St_a = pop(st_a)
  Push(st_a, a)
```

```
  Ret = a + (r - q)
  St_p = pop(st_p)
  St_s = pop(st_s)
  Last = cp
```

```
  Start = false
```

```
Return ret
```

```
}
```

```

1  AlgoritmoIter(A, p, s)
2  cp = p
3  cs = s
4  st_p = st_s = st_a = NIL
5  last = NIL
6  start = true
7  WHILE start || st_p != NIL DO
8    IF start THEN
9      IF cp + 1 > cs THEN
10         q <- Z_l(A, cp, cs)
11         r <- Z_r(A, cp, cs)
12         st_p = push(st_p, cp)
13         st_s = push(st_s, cs)
14         cs = q
15         /*start = true*/
16       ELSE /*caso base cs <= cp + 1*/
17         ret = 0
18         last = cp
19         start = false
20     ELSE /*ripresa del contesto*/
21       cp = top(st_p)
22       cs = top(st_s)
23       q <- Z_l(A, cp, cs)
24       r <- Z_r(A, cp, cs)
25       IF last = cp THEN
26         a <- ret
27         cp = q
28         cs = r
29         start = true
30       ELSE IF last = q THEN
31         a <- a - ret
32         cp = r
33         /*cs = cs*/
34         start = true
35       ELSE /*last = r*/
36         a <- a + ret
37         ret = a + (r - q)
38         st_p = pop(st_p)
39         st_s = pop(st_s)
40         last = cp
41         /*start = false*/
42   RETURN ret

```

a <- ret
push(st_a, a)

a = top(st_a)
a <- a - ret
st_a = pop(st_a)
push(st_a, a)

a = top(st_a)
a <- a + ret
st_a = pop(st_a)

1. Si risolva la seguente equazione di ricorrenza, calcolandone l'andamento asintotico:

$$T(n) = \begin{cases} 1, & \text{se } n \leq 27; \\ 3n^2 \cdot T(\sqrt[3]{n}) + 2n^3, & \text{altrimenti.} \end{cases}$$

2. Si scriva un **algoritmo iterativo** che **simuli precisamente** l'algoritmo ricorsivo di seguito riportato, dove Z_l e Z_r sono due funzioni esterne non meglio specificate che soddisfano la seguente proprietà: $p < Z_l(A, p, s) < Z_r(A, p, s) \leq s$ quando $p + 1 < s$.

function Algoritmo(A, p, s)

```

1  if s ≤ p + 1 then
2    return 0
3  else
4    q ← Z_l(A, p, s)
5    r ← Z_r(A, p, s)
6    a ← Algoritmo(A, p, q)
7    a ← a - Algoritmo(A, q, r)
8    a ← a + Algoritmo(A, r, s)
9    return a + (r - q)

```

ALGO_IT(A,p,s)

While(){

If(){ // NUOVA CHIAMATA
/* SIMULO FINO ALLA 1 CHIAMATA */

}else{ // VECCHIA CHIAMATA
/* RECUPERO CONTESTO */

If() { //torno 1 CHIAMATA con 2 CHIAMATA !=
da NULL

}else{ // 2 CHIAMATA

If(){ // torno dalla 1 CHIAMATA con 2
CHIAMATA = NULL

}else{ //TORNO DALLA 2 CHIAMATA
If(){ //torno dalla 2 CHIAMATA con 3
CHIAMATA != NULL

}else{ //TORNO dalla 3 CHIAMATA

}

}

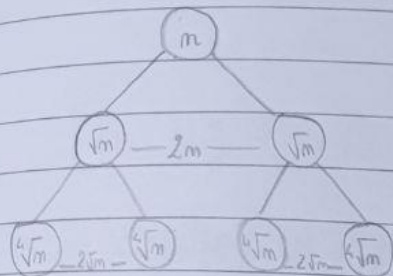
```
}
```

```
Return
```

```
}
```

EQUAZIONE 1

$$T(m) = \begin{cases} 1 & \text{se } m \leq 2 \\ 2m T(\sqrt{m}) + m^2 & \text{se } m > 2 \end{cases}$$



LIV.	ISTANZA	N° NODI	CONTR. NODO	CONTR. LIV.
0	m	1	m^2	m^2
1	$\sqrt{m}$	2m	m	$2m^2$
2	$\sqrt[4]{m}$	$2m \cdot 2\sqrt{m}$	$\sqrt{m}$	$2m \cdot 2\sqrt{m} \cdot \sqrt{m} = 4m^2$
i	$2^i \sqrt{m}$	$2m \cdot 2\sqrt{m} \cdot 2^i \sqrt{m}$	$(2^i \sqrt{m})^2$	$2^i m^2$

Calcolo $h := 2^i \sqrt{m} = 2 \Leftrightarrow m^{\frac{1}{2^i}} = 2 \Rightarrow \frac{1}{2^i} \log_m 2 \Leftrightarrow 2^i = \frac{1}{\log_m 2} \Rightarrow i = \log_2 (\log_2 m)$

Calcoliamo la sommatoria

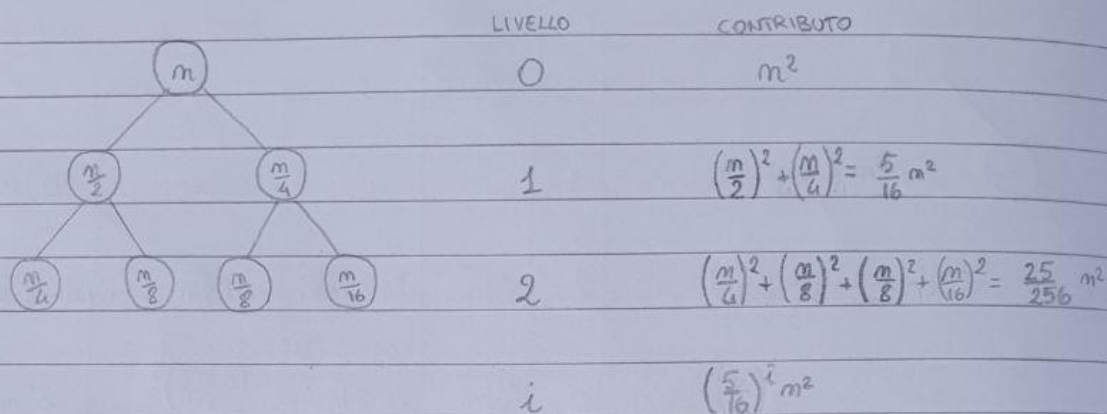
$$T(m) = \sum_{i=0}^h 2^i \cdot m^2 = \sum_{i=0}^{\log_2(\log_2 m)} 2^i \cdot m^2 \Rightarrow m^2 \cdot \sum_{i=0}^{\log_2(\log_2 m)} 2^i \quad \text{serie geometrica con ragione } > 1$$

$$= m^2 \cdot \left(\frac{2^{\log_2 \log_2 m + 1} - 1}{2 - 1} \right) = m^2 (2^{\log_2 \log_2 m} \cdot 2 - 1) = m^2 \cdot (2 \log_2 m - 1) = 2m^2 \log_2 m - m^2$$

$$= \Theta(m^2 \log_2 m)$$

EQUAZIONE 2

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ T\left(\frac{m}{2}\right) + T\left(\frac{m}{4}\right) + m^2 & \text{se } m > 1 \end{cases}$$



Calcolo $h_1 := \frac{n}{2^i} = 1 \Leftrightarrow 2^i = m \Rightarrow i = \log_2 m$

Calcolo $h_2 := \frac{n}{4^i} = 1 \Leftrightarrow 4^i = m \Rightarrow i = \log_4 m$

Calcolo la sommatoria:

$$T_{h_1}(m) = \sum_{i=0}^{h_1} \left(\frac{5}{16}\right)^i \cdot m^2 = \sum_{i=0}^{\log_2 m} \left(\frac{5}{16}\right)^i \cdot m^2 = m^2 \sum_{i=0}^{\log_2 m} \left(\frac{5}{16}\right)^i \quad \text{serie geometrica con ragione } < 1$$

$$= m^2 \cdot \frac{1}{1 - \frac{5}{16}} = m^2 \cdot \frac{1}{\frac{11}{16}} = 4 m^2 = \Theta(m^2)$$

Calcolo la sommatoria:

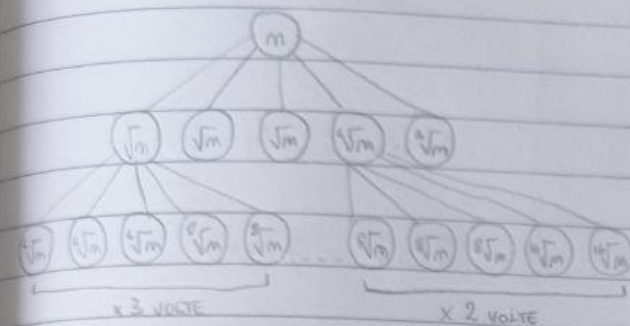
$$T_{h_2}(m) = \sum_{i=0}^{h_2} \left(\frac{5}{16}\right)^i \cdot m^2 = \sum_{i=0}^{\log_4 m} \left(\frac{5}{16}\right)^i \cdot m^2 = m^2 \cdot \sum_{i=0}^{\log_4 m} \left(\frac{5}{16}\right)^i \quad \text{serie geometrica con ragione } < 1$$

$$= m^2 \cdot \frac{1}{1 - \frac{5}{16}} = m^2 \cdot \frac{1}{\frac{11}{16}} = 4 m^2 = \Theta(m^2)$$

Sapendo che $T_{h_1}(m) \leq T(m) \leq T_{h_2}(m) \Rightarrow \Theta(m^2) \leq T(m) \leq \Theta(m^2) \Rightarrow T(m) = \Theta(m^2)$

EQUAZIONE 3

$$T(m) = \begin{cases} 1 & \text{se } m \leq 2 \\ 3T(\sqrt{m}) + 2T(\sqrt[4]{m}) + \log m & \text{se } m > 2 \end{cases}$$



LIVELLO	CONTRIBUTO
0	$\log m$
1	$3 \log m^{\frac{1}{2}} + 2 \log m^{\frac{1}{4}} = (\frac{3}{2} + \frac{1}{2}) \log m = 2 \log m$
2	$3(3 \log m^{\frac{1}{4}} + 2 \log m^{\frac{1}{8}}) + 2(3 \log m^{\frac{1}{8}} + 2 \log m^{\frac{1}{16}}) =$ $= (\frac{9}{4} + \frac{3}{2}) \log m + (\frac{3}{2} + \frac{1}{2}) \log m = 3 \log m + \log m = 4 \log m$
i	$2^i \log m$

Calcolo $h_1: \sqrt[2]{m} = 2 \Leftrightarrow m^{\frac{1}{2}} = 2 \Leftrightarrow \frac{1}{2} = \log_m 2 \Leftrightarrow 2^i = \log_2 m \Rightarrow i = \log_2(\log_2 m)$

Calcolo $h_2: \sqrt[4]{m} = 2 \Leftrightarrow m^{\frac{1}{4}} = 2 \Leftrightarrow \frac{1}{4} = \log_m 2 \Leftrightarrow 4^i = \log_4(\log_2 m) = \frac{\log_2(\log_2 m)}{2}$

Calcolo la sommatoria

$$T_{h_1}(m) = \sum_{i=0}^{h_1} 2^i \cdot \log m = \sum_{i=0}^{\log_2(\log_2 m)} 2^i \cdot \log m = \log m \cdot \sum_{i=0}^{\log_2(\log_2 m)} 2^i \text{ serie geometrica con ragione } > 1$$

$$= \log m \cdot \left(\frac{2^{\log_2(\log_2 m) + 1} - 1}{2 - 1} \right) = \log m (2^{\log_2(\log_2 m)} \cdot 2 - 1) = (2 \log_2 m - 1) \cdot \log m = 2 \log^2 m - \log m = \Theta(\log^2 m)$$

Calcolo la sommatoria:

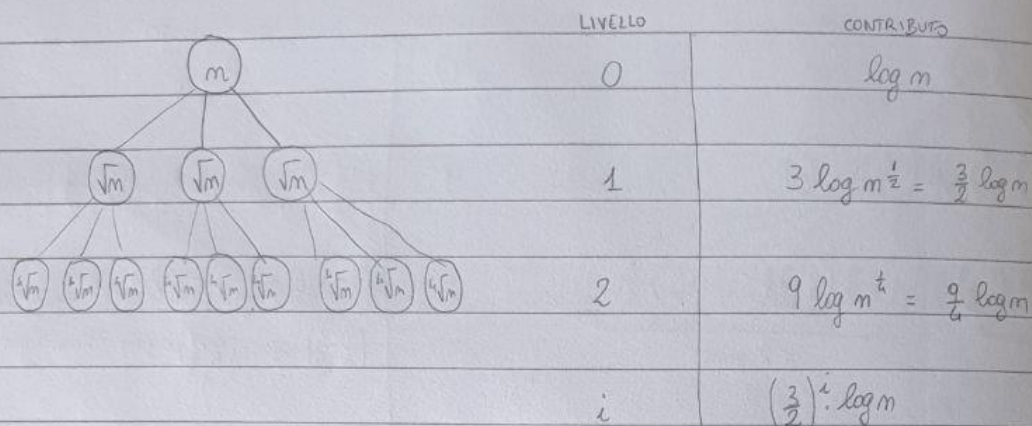
$$T_{h_2}(m) = \sum_{i=0}^{h_2} 2^i \cdot \log m = \sum_{i=0}^{\log_4(\log_2 m)} 2^i \cdot \log m = \log m \cdot \sum_{i=0}^{\log_4(\log_2 m)} 2^i \text{ serie geometrica con ragione } > 1$$

$$= \log m \cdot \left(\frac{2^{\log_4(\log_2 m) + 1} - 1}{2 - 1} \right) = \log m (2^{\frac{\log_2(\log_2 m)}{2}} \cdot 2 - 1) = \log m (2^{\frac{\log_2(\log_2 m)}{2}} \cdot 2 - 1) = \log m (2^{\frac{\log_2(\log_2 m)}{2} + 1} - 1)$$

$$= \log m \left[(2 \log_2 m)^{\frac{1}{2}} - 1 \right] = 2 (\log_2 m)^{\frac{3}{2}} - \log_2 m = \Theta(\log^{\frac{3}{2}}(m)) \Rightarrow \Theta(\log^2 m) \leq T(m) \leq \Theta(\log^{\frac{3}{2}}(m))$$

EQUAZIONE 4

$$T(m) = \begin{cases} 1 & \text{se } m \leq 2 \\ 3T(\sqrt{m}) + \log m & \text{se } m > 2 \end{cases}$$



Calcolo h: $2^i \sqrt{m} = 2 \Leftrightarrow m^{\frac{1}{2^i}} = 2 \Leftrightarrow \frac{1}{2^i} = \log_m 2 \Rightarrow i = \log_2 (\log_2 m)$

Calcolo la sommatoria:

$$T(m) = \sum_{i=0}^h \left(\frac{3}{2}\right)^i \cdot \log m = \sum_{i=0}^{\log_2 \log_2 m} \left(\frac{3}{2}\right)^i \cdot \log m = \log m \cdot \sum_{i=0}^{\log_2 \log_2 m} \left(\frac{3}{2}\right)^i \text{ serie geometrica con ragione } > 1$$

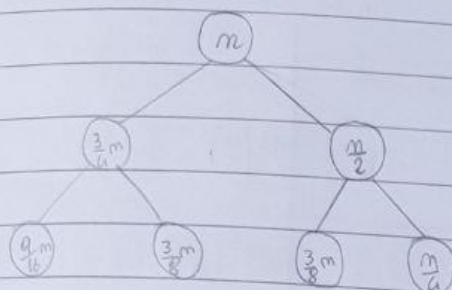
$$= \log m \cdot \frac{\left(\frac{3}{2}\right)^{\log_2 (\log_2 m) + 1} - 1}{\frac{3}{2} - 1} = \log m \cdot 2 \left(\left(\frac{3}{2}\right)^{\log_2 \log_2 m} \cdot \frac{3}{2} - 1 \right) = 3 \log m \cdot \left(\frac{3}{2}\right)^{\log_2 \log_2 m} - 2 \log m$$

$$= 3 \log m \cdot (\log_2 m)^{\log_2 \left(\frac{3}{2}\right)} - 2 \log m = 3 \log m \cdot (\log_2 m)^{\log_2(3) - 1} - 2 \log m$$

$$= 3 \log m \cdot \frac{(\log_2 m)^{\log_2(3)}}{\log_2 m} - 2 \log m = 3 (\log_2 m)^{\log_2(3)} - 2 \log m \Rightarrow \Theta((\log_2 m)^{\log_2(3)})$$

EQUAZIONE 5

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ T(\frac{3}{4}m) + T(\frac{1}{2}m) + m & \text{se } m > 1 \end{cases}$$



LIVELLO

CONTRIBUTO

0

m

1

$$\frac{3}{4}m + \frac{1}{2}m = \frac{5}{4}m$$

2

$$\frac{9}{16}m + \frac{3}{8}m + \frac{3}{8}m + \frac{1}{4}m = \frac{25}{16}m = \left(\frac{5}{4}\right)^2 m$$

i

$$\left(\frac{5}{4}\right)^i m$$

$$\text{Calcolo } h_1 := \left(\frac{3}{4}\right)^i m = 1 \Rightarrow \left(\frac{4}{3}\right)^i = m \Rightarrow i = \log_{\frac{4}{3}} m$$

$$\text{Calcolo } h_2 := \frac{m}{2^i} = 1 \Rightarrow 2^i = m \Rightarrow i = \log_2 m$$

Calcolo la sommatoria:

$$T_{h_1}(m) = \sum_{i=0}^{h_1} \left(\frac{5}{4}\right)^i m = \sum_{i=0}^{\log_{\frac{4}{3}} m} \left(\frac{5}{4}\right)^i m = m \cdot \sum_{i=0}^{\log_{\frac{4}{3}} m} \left(\frac{5}{4}\right)^i \quad \text{serie geometrica con ragione } > 1$$

$$= m \frac{\left(\frac{5}{4}\right)^{\log_{\frac{4}{3}} m + 1} - 1}{\frac{5}{4} - 1} = m \cdot 4 \left(\left(\frac{5}{4}\right)^{\log_{\frac{4}{3}} m} \cdot \frac{5}{4} - 1 \right) = 5m \cdot \left(\frac{5}{4}\right)^{\log_{\frac{4}{3}} m} - 4m = 5m \cdot m^{\log_{\frac{4}{3}} \frac{5}{4}} - 4m$$

$$= 5 \cdot m^{\log_{\frac{4}{3}} \left(\frac{5}{4}\right) + 1} - 4m \Rightarrow \Theta(m^{\log_{\frac{4}{3}} \left(\frac{5}{4}\right) + 1})$$

Calcolo la sommatoria:

$$T_{h_2}(m) = \sum_{i=0}^{h_2} \left(\frac{5}{4}\right)^i m = \sum_{i=0}^{\log_2 m} \left(\frac{5}{4}\right)^i m = m \cdot \sum_{i=0}^{\log_2 m} \left(\frac{5}{4}\right)^i \quad \text{serie geometrica con ragione } > 1$$

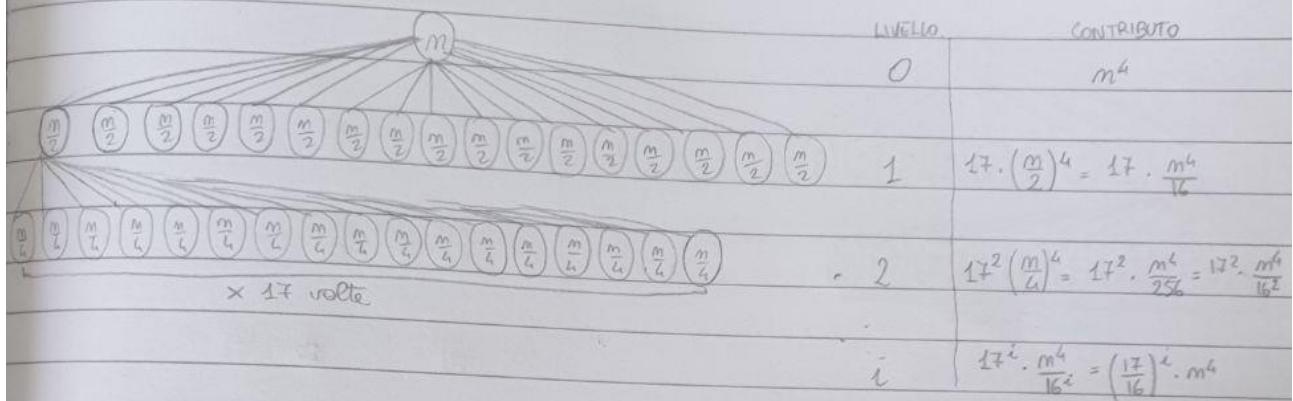
$$= 5m \left(\frac{5}{4}\right)^{\log_2 m} - 4m = 5m \cdot m^{\log_2 \left(\frac{5}{4}\right)} - 4m = 5m^{\log_2 \left(\frac{5}{4}\right) + 1} - 4m \Rightarrow \Theta(m^{\log_2 \left(\frac{5}{4}\right) + 1})$$

$$\text{Essendo } T_{h_2}(m) \leq T(m) \leq T_{h_1}(m) \text{ avremo } \Theta(m^{\log_2 \left(\frac{5}{4}\right) + 1}) \leq T(m) \leq \Theta(m^{\log_{\frac{4}{3}} \left(\frac{5}{4}\right) + 1})$$

N.B. Abbiamo sotto che $T_{h_2}(m) \leq T(m) \leq T_{h_1}(m)$ poiché $T_{h_2}(m)$ decresce più velocemente e quindi termina prima di $T_{h_1}(m)$

EQUAZIONE 7

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ 17T\left(\frac{m}{2}\right) + m^4 & \text{se } m > 1 \end{cases}$$



Calcolo $h := \frac{m}{2^i} = 1 \Rightarrow 2^i = m \Rightarrow i = \log_2 m$

Calcolo la sommatoria:

$$T(m) = \sum_{i=0}^h \left(\frac{17}{16}\right)^i \cdot m^4 = \sum_{i=0}^{\log_2 m} \left(\frac{17}{16}\right)^i \cdot m^4 = m^4 \cdot \sum_{i=0}^{\log_2 m} \left(\frac{17}{16}\right)^i \quad \text{serie geometrica con ragione } > 1$$

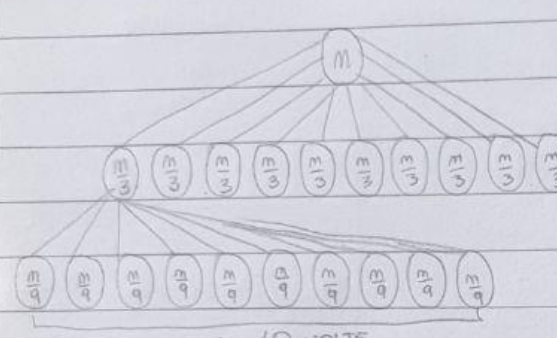
$$= m^4 \cdot \frac{\left(\frac{17}{16}\right)^{h+1} - 1}{\frac{17}{16} - 1} = m^4 \cdot \frac{\left(\frac{17}{16}\right)^{\log_2 m + 1} - 1}{\frac{1}{16}} = m^4 \left[16 \left(\left(\frac{17}{16}\right)^{\log_2 m} \cdot \frac{17}{16} - 1 \right) \right]$$

$$= 17 m^4 \cdot \left(\frac{17}{16}\right)^{\log_2 m} - 16 m^4 = 17 m^4 \cdot m^{\log_2 \left(\frac{17}{16}\right)} - 16 m^4 = 17 m^{\log_2 \left(\frac{17}{16}\right) + 4} - 16 m^4$$

$$\Rightarrow T(m) = \Theta\left(m^{\log_2 \left(\frac{17}{16}\right) + 4}\right)$$

EQUAZIONE 8

$$T(m) = \begin{cases} 1 & \text{se } m \leq 2 \\ 10T\left(\frac{m}{3}\right) + m^2 & \text{se } m > 2 \end{cases}$$

	LIVELLO	CONTRIBUTO
	0	m^2
	1	$10 \left(\frac{m}{3}\right)^2 = 10 \cdot \frac{m^2}{9}$
	2	$10^2 \left(\frac{m}{9}\right)^2 = 10^2 \cdot \frac{m^2}{81}$
	i	$10^i \cdot \left(\frac{m}{3^i}\right)^2 = 10^i \cdot \frac{m^2}{9^i} = \left(\frac{10}{9}\right)^i m^2$

Calcolo $h := \frac{m}{3^i} = 2 \Rightarrow 3^i = \frac{m}{2} \Rightarrow i = \log_3\left(\frac{m}{2}\right)$

Calcolo la sommatoria:

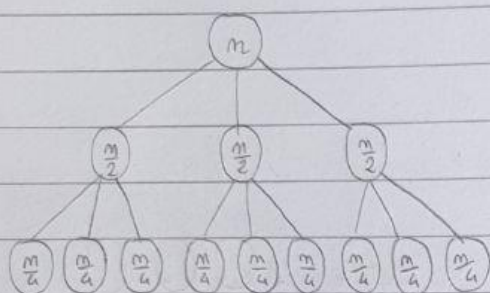
$$T(m) = \sum_{i=0}^h \left(\frac{10}{9}\right)^i \cdot m^2 = \sum_{i=0}^{\log_3(\frac{m}{2})} \left(\frac{10}{9}\right)^i \cdot m^2 = m^2 \cdot \sum_{i=0}^{\log_3(\frac{m}{2})} \left(\frac{10}{9}\right)^i \quad \text{serie geometrica con ragione } > 1$$

$$= m^2 \cdot \frac{\left(\frac{10}{9}\right)^{h+1} - 1}{\frac{10}{9} - 1} = m^2 \cdot \frac{\left(\frac{10}{9}\right)^{\log_3(\frac{m}{2})+1} - 1}{\frac{1}{9}} = m^2 \left[9 \left(\frac{10}{9}\right)^{\log_3(\frac{m}{2})} \cdot \frac{10}{9} - 1 \right]$$

$$= m^2 \left[\left(\frac{10}{9}\right)^{\log_3(\frac{m}{2})} \cdot 10 - 9 \right] = 10 m^2 \cdot \left(\frac{10}{9}\right)^{\log_3(\frac{m}{2})} - 9 m^2 = 10 m^2 \cdot \left(\frac{m}{2}\right)^{\log_3(\frac{10}{9})} - 9 m^2$$

EQUAZIONE 10

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ 3T\left(\frac{m}{2}\right) + m & \text{se } m > 1 \end{cases}$$

	LIVELLO	CONTRIBUTO
	0	m
	1	$\frac{m}{2} + \frac{m}{2} + \frac{m}{2} = \frac{3}{2}m$
	2	$9\left(\frac{m}{4}\right) = \frac{9}{4}m = \left(\frac{3}{2}\right)^2 m$
	i	$\left(\frac{3}{2}\right)^i \cdot m$

Calcolo h : $\frac{m}{2^i} = 1 \Rightarrow 2^i = m \Rightarrow i = \log_2 m$

Calcolo la sommatoria:

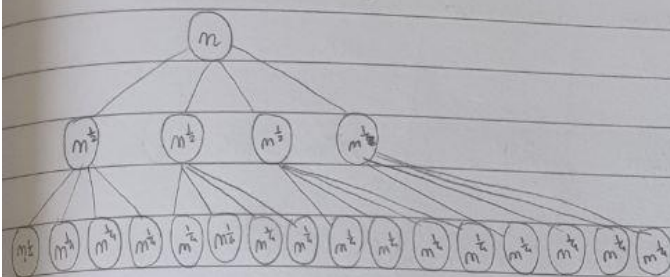
$$T(m) = \sum_{i=0}^h \left(\frac{3}{2}\right)^i \cdot m = \sum_{i=0}^{\log_2 m} \left(\frac{3}{2}\right)^i \cdot m = m \cdot \sum_{i=0}^{\log_2 m} \left(\frac{3}{2}\right)^i \text{ serie geometrica con ragione } > 1$$

$$= m \cdot \frac{\left(\frac{3}{2}\right)^{h+1} - 1}{\frac{3}{2} - 1} = m \left[2 \left(\frac{3}{2} \cdot \left(\frac{3}{2}\right)^{\log_2 m} - 1 \right) \right] = 3m \left(\frac{3}{2}\right)^{\log_2 m} - 2m$$

$$= 3m \cdot m^{\log_2\left(\frac{3}{2}\right)} - 2m = 3m^{\log_2\left(\frac{3}{2}\right)+1} - 2m \Rightarrow T(m) = \Theta\left(m^{\log_2\left(\frac{3}{2}\right)+1}\right)$$

EQUAZIONE 11

$$T(m) = \begin{cases} 1 & \text{se } m \leq 2 \\ 4T(\sqrt{m}) + \log m & \text{se } m > 2 \end{cases}$$



LIVELLO

CONTRIBUTO

0

$\log m$

1

$4 \log \sqrt{m} = 4 \log m^{1/2}$

2

$16 \log \sqrt[4]{m} = 4^2 \log m^{1/4}$

i

$4^i \log \sqrt[i]{m} = 4^i \log m^{1/i} = 2^i \cdot \log m$

Calcolo $h := m^{1/2^i} = 2 \Rightarrow \frac{1}{2^i} = \log_m 2 \Rightarrow 2^i = \log_2 m \Rightarrow i = \log_2 \log_2 m$

Calcolo la sommatoria:

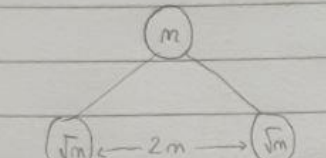
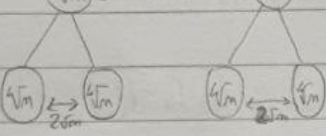
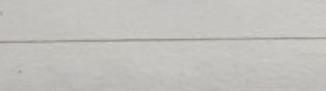
$$T(m) = \sum_{i=0}^h 2^i \cdot \log m = \sum_{i=0}^{\log_2 \log_2 m} 2^i \cdot \log m = \log m \cdot \sum_{i=0}^{\log_2 \log_2 m} 2^i \quad \text{serie geometrica con ragione } > 1$$

$$= \log m \cdot \frac{2^{h+1} - 1}{2 - 1} = \log m \cdot (2^{\log_2 \log_2 m + 1} - 1) = \log m (2^{\log_2 \log_2 m} \cdot 2 - 1)$$

$$= 2 \cdot \log m \cdot \log_2 m - \log m \Rightarrow T(m) = \Theta(\log^2 m)$$

EQUAZIONE 12

$$T(m) = \begin{cases} 1 & \text{se } m \leq 2 \\ 2mT(\sqrt{m}) + m^2 & \text{se } m > 2 \end{cases}$$

	LIVELLO	ISTANZA	N° NODI	CONTR. NODO	CONTR. LOCALE LIVELLO
	0	m	1	m^2	m^2
	1	$\sqrt{m}$	2	$(\sqrt{m})^2 = m$	$2m \cdot m = 2m^2$
	2	$\sqrt[4]{m}$	$2m \cdot 2\sqrt{m}$	$(\sqrt[4]{m})^2 = \sqrt{m}$	$2m \cdot 2\sqrt{m} \cdot \sqrt{m} = 4m^2 = 2^2 m^2$
	i	$2^i \sqrt{m}$	$2m \cdot 2\sqrt{m} \cdot \dots \cdot 2^i \sqrt{m}$	$(2^i \sqrt{m})^2$	$2^i \cdot m^2$

Calcolo h: $2^i \sqrt{m} = 2 \Rightarrow m^{\frac{1}{2} \cdot 2^i} = 2 \Rightarrow \frac{1}{2^i} = \log_m 2 \Rightarrow i = \log_2 (\log_2 m)$

Calcolo la sommatoria:

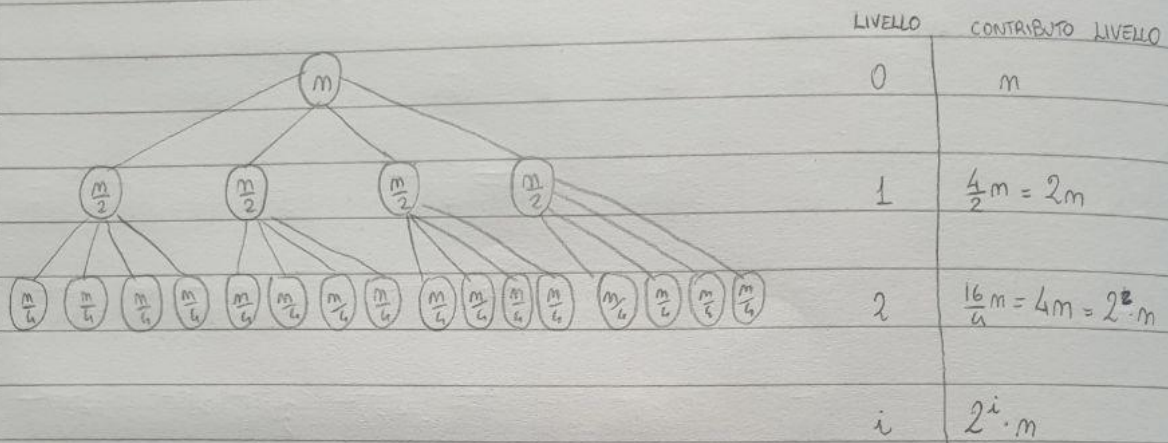
$$T(m) = \sum_{i=0}^h 2^i \cdot m^2 = \sum_{i=0}^{\log_2 \log_2 m} 2^i \cdot m^2 = m^2 \cdot \sum_{i=0}^{\log_2 \log_2 m} 2^i \quad \text{serie geometrica con ragione } > 1$$

$$= m^2 \cdot \frac{2^{h+1} - 1}{2 - 1} = m^2 \cdot (2^{\log_2 \log_2 m + 1} - 1) = m^2 \cdot (2^{\log_2 \log_2 m} \cdot 2 - 1)$$

$$= 2m^2 \log_2 m - m^2 \Rightarrow T(m) = \Theta(m^2 \log_2 m)$$

EQUAZIONE 14

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ 4T\left(\frac{m}{2}\right) + m & \text{se } m > 1 \end{cases}$$



Calcolo h: $\frac{m}{2^i} = 1 \Rightarrow 2^i = m \Rightarrow i = \log_2 m$

Calcolo la sommatoria:

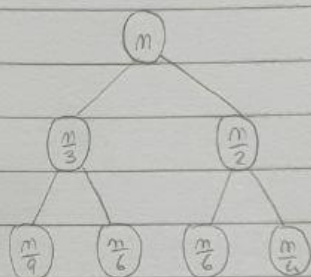
$$T(m) = \sum_{i=0}^h 2^i \cdot m = \sum_{i=0}^{\log_2 m} 2^i \cdot m = m \cdot \sum_{i=0}^{\log_2 m} 2^i \quad \text{serie geometrica con ragione } > 1$$

$$= m \cdot \frac{2^{h+1} - 1}{2 - 1} = m(2^{\log_2 m + 1} - 1) = m(2^{\log_2 m} \cdot 2 - 1) = m(m \cdot 2 - 1)$$

$$= 2m^2 - m \Rightarrow T(m) = \Theta(m^2)$$

EQUAZIONE 18

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ T\left(\frac{m}{3}\right) + T\left(\frac{m}{2}\right) + m & \text{se } m > 1 \end{cases}$$



LIVELLO CONTRIBUTO LIVELLO

0

m

1

$$\frac{m}{3} + \frac{m}{2} = \frac{2m+3m}{6} = \frac{5}{6}m$$

2

$$\frac{m}{9} + \frac{m}{6} + \frac{m}{6} + \frac{m}{4} = \frac{4m+6m+6m+9m}{36} = \frac{25}{36}m = \left(\frac{5}{6}\right)^2 m$$

i

$$\left(\frac{5}{6}\right)^i \cdot m$$

Calcolo $h_1 := \frac{m}{3^i} = 1 \Rightarrow 3^i = m \Rightarrow i = \log_3 m$

Calcolo $h_2 := \frac{m}{2^i} = 1 \Rightarrow 2^i = m \Rightarrow i = \log_2 m$

Calcolo la sommatoria:

$$T_{h_1}(m) = \sum_{i=0}^{h_1} \left(\frac{5}{6}\right)^i \cdot m = \sum_{i=0}^{\log_3 m} \left(\frac{5}{6}\right)^i \cdot m = m \cdot \sum_{i=0}^{\log_3 m} \left(\frac{5}{6}\right)^i \quad \text{serie geometrica con ragione } < 1$$

$$= m \cdot \frac{1}{1 - \frac{5}{6}} = m \cdot \frac{1}{\frac{1}{6}} = 6m \Rightarrow T_{h_1}(m) = \Theta(m)$$

Calcolo la sommatoria:

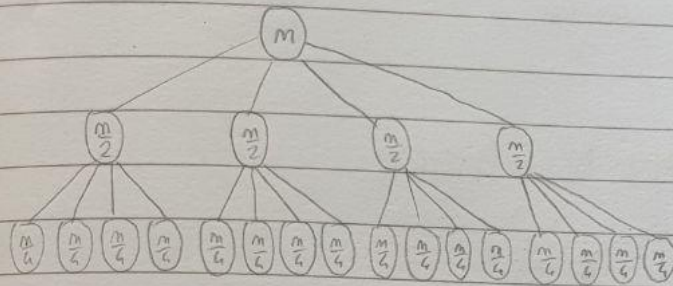
$$T_{h_2}(m) = \sum_{i=0}^{h_2} \left(\frac{5}{6}\right)^i \cdot m = \sum_{i=0}^{\log_2 m} \left(\frac{5}{6}\right)^i \cdot m = m \cdot \sum_{i=0}^{\log_2 m} \left(\frac{5}{6}\right)^i \quad \text{serie geometrica con ragione } < 1$$

$$= m \cdot \frac{1}{1 - \frac{5}{6}} = m \cdot \frac{1}{\frac{1}{6}} = 6m \Rightarrow T_{h_2}(m) = \Theta(m)$$

Essendo $T_{h_1}(m) \leq T(m) \leq T_{h_2}(m)$, avremo che $T(m) = \Theta(m)$

EQUAZIONE 19

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ 4T\left(\frac{m}{2}\right) + m^2 & \text{se } m > 1 \end{cases}$$



LIVELLO	CONTRIBUTO LIVELLO
0	m^2
1	$4\left(\frac{m}{2}\right)^2 = 4 \cdot \frac{m^2}{4} = m^2$
2	$16\left(\frac{m}{4}\right)^2 = 16 \cdot \frac{m^2}{16} = m^2$
i	$4^i \left(\frac{m}{2^i}\right)^2 = m^2$

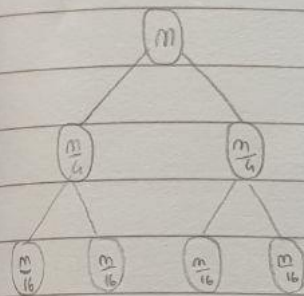
Calcolo $h := \frac{m}{2^i} = 1 \Rightarrow 2^i = m \Rightarrow i = \log_2 m$

Calcolo la sommatoria:

$$T(m) = \sum_{i=0}^h m^2 = \sum_{i=0}^{\log_2 m} m^2 = m^2 \cdot \sum_{i=0}^{\log_2 m} (1) \Rightarrow T(m) = \Theta(m^2)$$

EQUAZIONE 21

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ 2T\left(\frac{m}{4}\right) + \sqrt{m} & \text{se } m > 1 \end{cases}$$



LIVELLO	CONTRIBUTO LIVELLO
0	$\sqrt{m}$
1	$\sqrt{\frac{m}{4}} + \sqrt{\frac{m}{4}} = \frac{\sqrt{m}}{2} + \frac{\sqrt{m}}{2} = \sqrt{m}$
2	$4\sqrt{\frac{m}{16}} = 4 \cdot \frac{\sqrt{m}}{4} = \sqrt{m}$
i	$2^i \cdot \sqrt{\frac{m}{4^i}} = 2^i \cdot \frac{\sqrt{m}}{2^i} = \sqrt{m}$

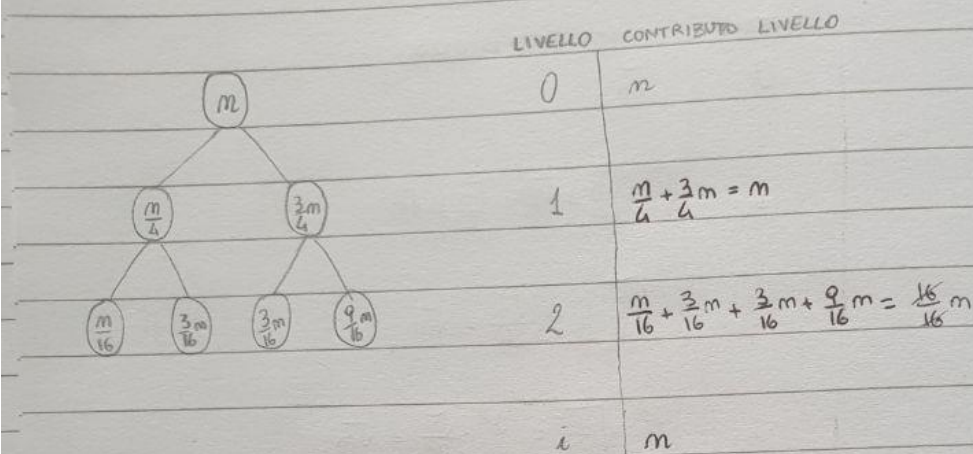
Calcolo $h := \frac{m}{4^i} = 1 \Rightarrow 4^i = m \Rightarrow \log_4 m$

Calcolo la sommatoria

$$T(m) = \sum_{i=0}^h \sqrt{m} = \sum_{i=0}^{\log_4 m} \sqrt{m} = \sqrt{m} \cdot \sum_{i=0}^{\log_4 m} 1 \Rightarrow T(m) = \Theta(\sqrt{m})$$

EQUAZIONE 22

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ T\left(\frac{m}{4}\right) + T\left(\frac{3m}{4}\right) + m & \text{se } m > 1 \end{cases}$$



Calcolo $h_1 := \frac{m}{4^i} = 1 \Rightarrow 4^i = m \Rightarrow i = \log_4 m$

Calcolo $h_2 := \left(\frac{3}{4}\right)^i m = 1 \Rightarrow m = \left(\frac{4}{3}\right)^i \Rightarrow i = \log_{\frac{4}{3}} m$

Calcolo la sommatoria

$$T_{h_1}(m) = \sum_{i=0}^{h_1} m^{\oplus} = \sum_{i=0}^{\log_4 m} m^{\oplus} = m^{\oplus} \cdot \sum_{i=0}^{\log_4 m} 1 \Rightarrow T_{h_1}(m) = \Theta(m^{\oplus})$$

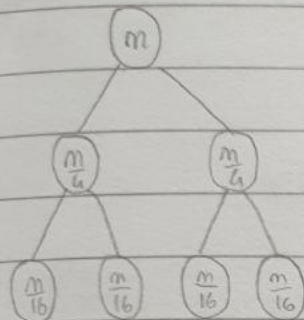
Calcolo la sommatoria

$$T_{h_2}(m) = \sum_{i=0}^{h_2} m^{\oplus} = \sum_{i=0}^{\log_{\frac{4}{3}} m} m^{\oplus} = m^{\oplus} \cdot \sum_{i=0}^{\log_{\frac{4}{3}} m} 1 \Rightarrow T_{h_2}(m) = \Theta(m^2)$$

Essendo $T_{h_1}(m) \leq T(m) \leq T_{h_2}(m)$, avremo che $T(m) = \Theta(m)$

EQUAZIONE 23

$$T(m) = \begin{cases} 1 & \text{se } m \leq 1 \\ 2T\left(\frac{m}{4}\right) + m^3 & \text{se } m > 1 \end{cases}$$



LIVELLO CONTRIBUTO LIVELLO

0	m^3
1	$2\left(\frac{m}{4}\right)^3 = 2 \cdot \frac{m^3}{64} = \frac{m^3}{32}$
2	$4\left(\frac{m}{16}\right)^3 = 4 \cdot \frac{m^3}{4096} = \frac{m^3}{1024}$
i	$\frac{m^3}{32^i} = \left(\frac{1}{32}\right)^i \cdot m^3$

Calcolo $h := \frac{m}{4^i} = 1 \Rightarrow 4^i = m \Rightarrow i = \log_4 m$

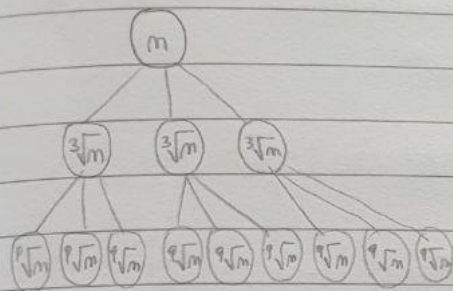
calcolo la sommatoria:

$$T(m) = \sum_{i=0}^h \left(\frac{1}{32}\right)^i \cdot m^3 = \sum_{i=0}^{\log_4 m} \left(\frac{1}{32}\right)^i \cdot m^3 = m^3 \cdot \sum_{i=0}^{\log_4 m} \left(\frac{1}{32}\right)^i \quad \text{serie geometrica con ragione } < 1$$

$$= m^3 \cdot \frac{1}{1 - \frac{1}{32}} = m^3 \cdot \frac{1}{\frac{31}{32}} = \frac{32}{31} m^3 \Rightarrow T(m) = \Theta(m^3)$$

EQUAZIONE 24

$$T(m) = \begin{cases} 1 & \text{se } m \leq 3 \\ 3T(\sqrt[3]{m}) + \log_3 m & \text{se } m > 3 \end{cases}$$



LIVELLO	CONTRIBUTO LIVELLO
0	$\log_3 m$
1	$3 \log_3 \sqrt[3]{m} = 3 \log_3 m^{\frac{1}{3}} = 3 \cdot \frac{1}{3} \log_3 m$
2	$3^2 \cdot \log_3 \sqrt[9]{m} = 3^2 \cdot \log_3 m^{\frac{1}{9}} = 3^2 \cdot \frac{1}{9} \log_3 m$
i	$\log_3 m$

Calcolo $h := \sqrt[3]{m} = 3 \Rightarrow m^{\frac{1}{3^i}} = 3 \Rightarrow \frac{1}{3^i} = \log_3 3 \Rightarrow 3^i = \frac{1}{\log_3 3} \Rightarrow 3^i = \log_3 m \Rightarrow i = \log_3 (\log_3 m)$

Calcolo la sommatoria:

$$T(m) = \sum_{i=0}^h \log_3 m = \sum_{i=0}^{\log_3 \log_3 m} \log_3 m = \log_3 m \cdot \sum_{i=0}^{\log_3 \log_3 m} 1 \Rightarrow T(m) = \Theta(\log_3 m)$$

EQUAZIONE 28

$$T(m) = \begin{cases} \text{costante} & \text{se } m \leq x \\ 3T(\sqrt{m}) + 2T(\sqrt[3]{m}) + \log_x m & \text{se } m > x \end{cases}$$

LIVELLO	CONTRIBUTO	LIVELLO
0	$\log_x m$	
1	$3 \log_x m^{\frac{1}{2}} + 2 \log_x m^{\frac{1}{3}} = \left(\frac{3}{2} + \frac{2}{3}\right) \log_x m = \frac{13}{6} \log_x m$	
2	$3(3m^{\frac{1}{4}} + 2m^{\frac{1}{6}}) + 2(3m^{\frac{1}{6}} + 2m^{\frac{1}{9}}) = \left(\frac{169}{36}\right) \log_x m = \left(\frac{13}{16}\right)^2 \log_x m$	
i	$\left(\frac{13}{16}\right)^i \cdot \log_x m$	

Calcolo $h_1 := \sqrt[2]{m} = x \Rightarrow m^{\frac{1}{2^i}} = x \Rightarrow \frac{1}{2^i} = \log_x m \Rightarrow 2^i = \log_x m \Rightarrow i = \log_2(\log_x m)$

Calcolo $h_2 := \sqrt[3]{m} = x \Rightarrow m^{\frac{1}{3^i}} = x \Rightarrow \frac{1}{3^i} = \log_x m \Rightarrow 3^i = \log_x m \Rightarrow i = \log_3(\log_x m)$

Calcolo la sommatoria:

$$T_{h_1}(m) = \sum_{i=0}^{h_1} \left(\frac{13}{16}\right)^i \cdot \log_x m = \sum_{i=0}^{\log_2(\log_x m)} \left(\frac{13}{16}\right)^i \cdot \log_x m = \log_x m \cdot \sum_{i=0}^{\log_2(\log_x m)} \left(\frac{13}{16}\right)^i \text{ serie geometrica con ragione } < 1$$

$$= \log_x m \cdot \frac{1}{1 - \frac{13}{16}} = \log_x m \cdot \left(\frac{16}{3}\right) = \frac{16}{3} \log_x m \Rightarrow T_{h_1}(m) = \Theta(\log_x m)$$

Calcolo la sommatoria:

$$T_{h_2}(m) = \sum_{i=0}^{h_2} \left(\frac{13}{16}\right)^i = \sum_{i=0}^{\log_3(\log_x m)} \left(\frac{13}{16}\right)^i \cdot \log_x m = \log_x m \cdot \sum_{i=0}^{\log_3(\log_x m)} \left(\frac{13}{16}\right)^i \text{ serie geometrica con ragione } < 1$$

$$= \log_x m \cdot \frac{1}{1 - \frac{13}{16}} = \frac{16}{3} \log_x m \Rightarrow T_{h_2}(m) = \Theta(\log_x m)$$

Essendo $T_{h_1}(m) \leq T(m) \leq T_{h_2}(m)$, avremo che $T(m) = \Theta(\log_x m)$